Analyzing Smart City Bike Sharing Data using Power BI

Project Overview:

- The Bike Sharing Analysis Project focuses on analyzing bike station data to understand usage patterns, revenue performance, and operational efficiency across different cities. Using Power BI, the dataset was cleaned and transformed, and an interactive dashboard was created to visualize key metrics such as bike availability, distance traveled, revenue generation, and cost efficiency.
- The analysis helps identify high-performing cities, revenue trends over time, and underperforming locations. These insights support better decision-making for improving bike distribution, optimizing operational costs, and enhancing the overall bike-sharing service.

Project Objectives:

- Analyze bike-sharing data to understand station performance and bike availability.
- Calculate important metrics such as revenue, distance traveled, revenue per km, and cost efficiency.
- Identify the cities with the highest and lowest revenue per km to evaluate operational performance.
- Examine revenue trends over the years to understand growth patterns in bike usage.
- Detects underperforming cities with low revenue efficiency and high operational costs.
- Develop an interactive Power BI dashboard to visualize insights and support data-driven decision-making.
- The objective of this project is to analyze bike-sharing station data using Power BI to identify usage patterns, revenue performance, and operational efficiency across cities. The project aims to clean and transform raw data, build a structured data model, create DAX measures, and design interactive dashboards to answer key business questions such as:

- Which cities generate the highest and lowest revenue per kilometer.
- How revenue trends change over time.
- Which cities or stations are underperforming with low cost efficiency.
- How bike availability and station occupancy affect system utilization.

Tool and Technologies:

- **Excel & Power Query -** Data Cleaning & Transformation
- **Power BI** - Data Visualization and dashboard development
- **DAX(Data Analysis Expression) -** Calculations and measures

Data Cleaning & Transformation:

Rename Columns

Old Name	**New Name**
number	Station_ID
name	Station_Name
address	Address
Contract Name	City
Bike Stands	Total_Bike_Stands
Available Bike Stands	Available_Stands
Available Bikes	Available_Bikes
Last Update date	Last_Update Date/time

Data Types

Correct data types improve analysis

Column	**Data Type**
Station_ID	Whole Number
Station_Name	Text
Address	Text
Banking.	True/False
Bonus	True/False
Status	Text
City.	Text
Total_Bike_Stands	Whole Number
Available_Stands	Whole Number
Available_Bikes	Whole Number
Last_Update	Date/Time

Split Date and Time
The Last Update column contains both date and time.
Steps:
Select Last_Update column.
Go to Transform → Split Column → By Delimiter.
Result:
Last_Update_Date
Last_Update_Time

Handle Missing Values
Select column.
Go to Transform → Replace Values.
Example: Replace null → 0 in numeric columns.

Remove Duplicate Records
Select Station_ID column.
Go to Home → Remove Rows → Remove Duplicates.

Filter Active Stations
Click the filter icon in the Status column.
Select OPEN only.

Create Calculated Column (Occupancy)
Go to Add Column → Custom Column.
Formula:

Occupancy Rate = [Available Bikes] / [Total_Bike_Stands]

Untitled - Power Query Editor

File | Home | Transform | Add Column | View | Tools | Help

Queries [1]

Feuil1

```
= Table.TransformColumnTypes(#"Added Custom2",{{"Occupancy Rate", Percentage.Type}})
```

		Available Bike Stands	Available Bikes	Last Update.1	Last Update.2	Occupancy Rate
1	1	0	0	02-06-2022	15:14:38	-100.00%
2	9	0	0	06-04-2022	12:54:35	-900.00%
3	15	0	0	27-09-2022	10:57:48	-1500.00%
4	15	15	0	08-06-2022	11:50:13	-1500.00%
5	0	0	0	23-05-2022	20:11:50	0.00%
6	10	0	0	13-09-2022	11:28:01	-1000.00%
7	24	24	0	06-09-2022	18:02:17	-2400.00%
8	20	13	7	14-09-2022	13:27:09	-1300.00%
9	10	0	0	12-09-2022	11:18:28	-1000.00%
10	20	2	0	03-10-2022	11:34:01	-2000.00%
11	10	10	0	13-09-2022	15:41:24	-1000.00%
12	14	0	0	26-09-2022	16:44:57	-1400.00%
13	15	0	0	03-10-2022	12:02:43	-1500.00%
14	10	0	0	10-10-2022	12:40:37	-1000.00%
15	8	0	0	26-09-2022	11:29:34	-800.00%
16	32	0	0	17-10-2022	11:36:42	-3200.00%
17	16	0	0	24-10-2022	13:08:08	-1600.00%
18	17	0	0	04-10-2022	11:49:03	-1700.00%
19	1	0	1	30-08-2022	21:00:02	0.00%
20	10	0	0	08-11-2022	14:53:04	-1000.00%
21	14	0	0	22-11-2022	12:47:11	-1400.00%
22	17	0	0	11-10-2022	11:40:16	-1700.00%
23	1	1	0	28-11-2022	14:00:36	-100.00%
24	15	0	0	22-11-2022	14:52:19	-1500.00%
25	12	0	0	24-10-2022	13:58:26	-1200.00%
26	19	0	0	29-11-2022	13:37:33	-1900.00%
27	14	13	0	19-12-2022	15:39:35	-1400.00%
28						

Query Settings

PROPERTIES

Name: Feuil1

All Properties

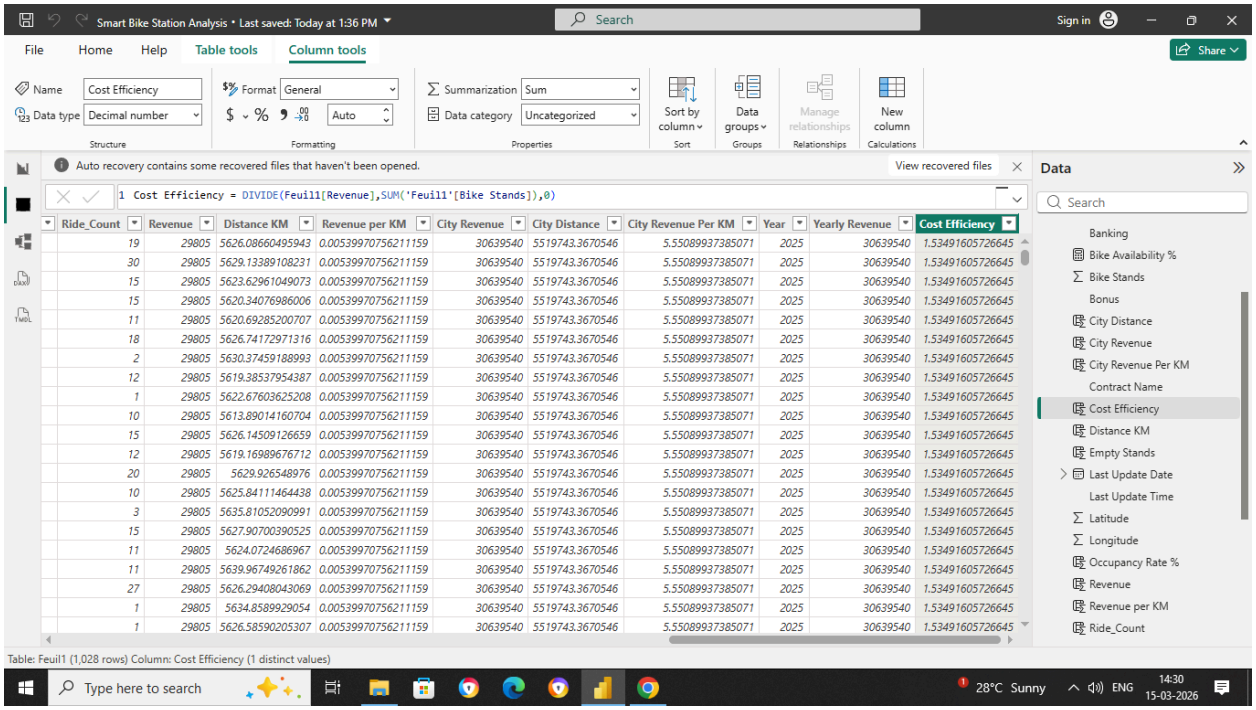
APPLIED STEPS

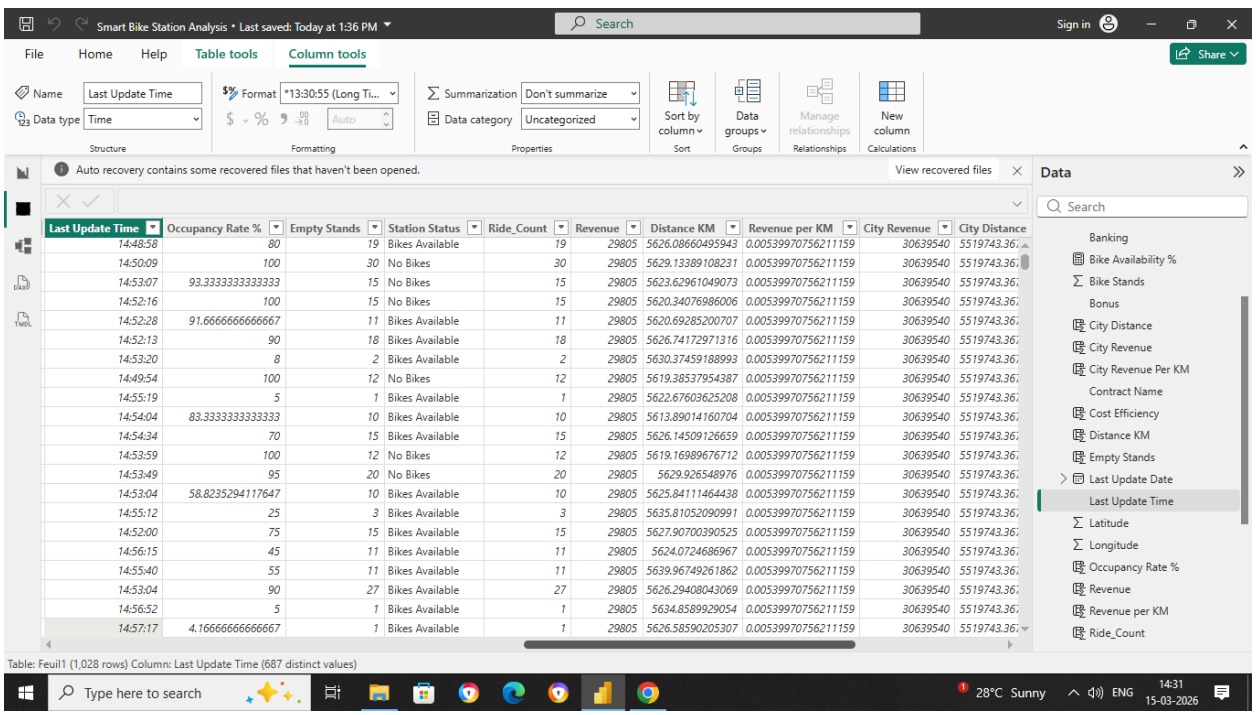
- Capitalized Each Word3
- Cleaned Text8
- Trimmed Text7
- Cleaned Text9
- Trimmed Text8
- Cleaned Text10
- Trimmed Text9
- Cleaned Text11
- Trimmed Text10
- Demoted Headers
- Changed Type1
- Promoted Headers1
- Changed Type2
- Renamed Columns
- Split Column by Delimiter
- Changed Type4
- Replaced Value
- Removed Duplicates1
- Added Custom
- Changed Type3
- Removed Columns
- Added Custom2
- Changed Type5

14 COLUMNS, 199+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 16:47

Type here to search 29°C Partly cloudy ENG 18:44 12-03-2026





Set Up a Fact Table

The Fact Table stores all measurable values such as bike counts and occupancy rate.

Fact Table: Fact_BikeSharing

Create Dimension Tables
Dimension tables contain descriptive information.

Dim_Station

Column
StationID
Station Name
Address
Banking
Bonus
Status

Dim_Location

Column
StationID
Latitude
Longitude
Contract Name

Dim_Date

Column
Date
Year
Month
Day

Cardinality

Dim_Station
Fact_BikeSharing
StationID One-to-Many (1:*)
Dim_Location
Fact_BikeSharing
StationID One-to-Many (1:*)
Dim_Date
Fact_BikeSharing
Date One-to-Many (1:*)

Cardinality Used

One-to-Many (1:*)

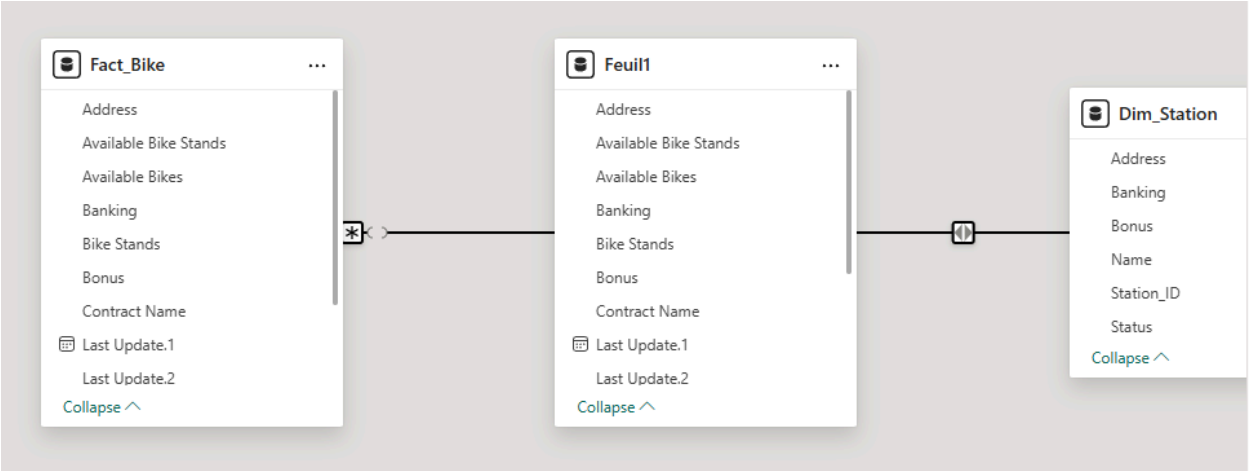
Example:

Dim_Station (1) → Fact_BikeSharing (*)

Explanation:

One station appears once in the dimension table.

The same station can appear many times in the fact table.



Calculated Columns (DAX)

Calculated columns are created using Model view → New Column.

Occupancy Rate Column

Occupancy Rate = DIVIDE('BikeData'[Available Bikes],'BikeData'[Bike Stands])

If you want percentage:

Occupancy Rate % = DIVIDE('BikeData'[Available Bikes],'BikeData'[Bike Stands]) * 100

Empty Stands Column

Empty Stands ='BikeData'[Bike Stands] - 'BikeData'[Available Bikes]

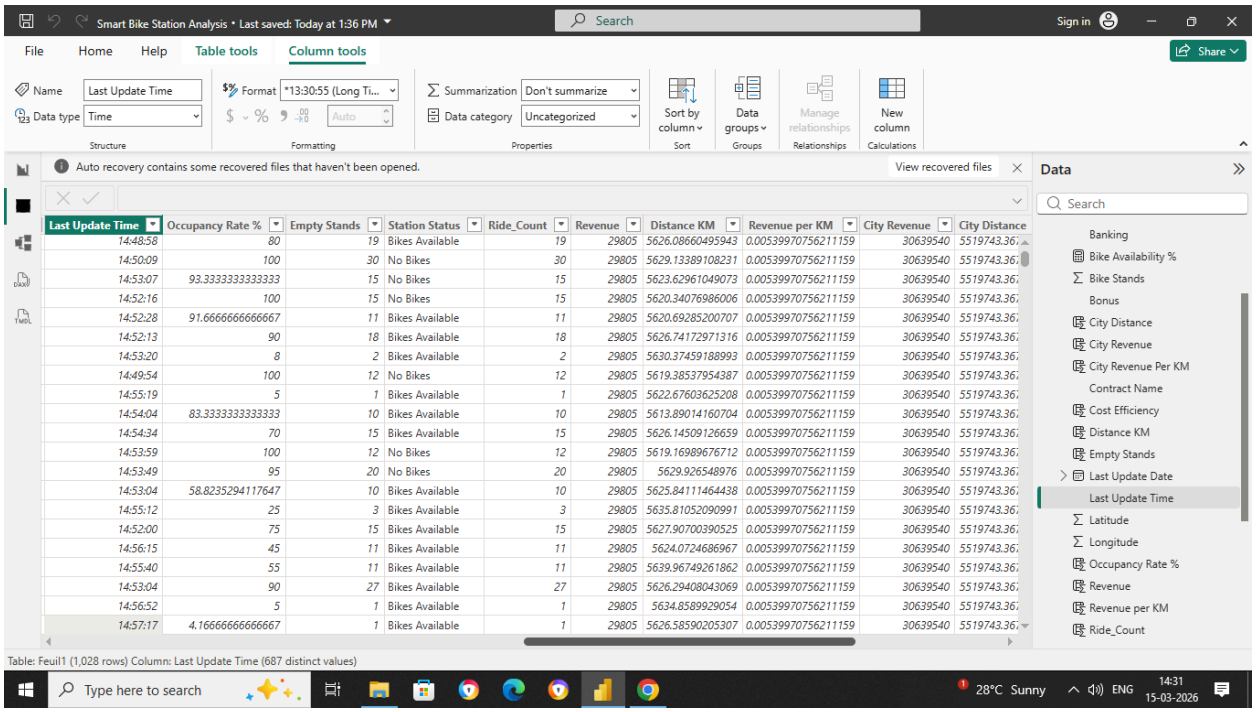
Station Status Column

Categorize stations.

Station Status =IF('BikeData'[Available Bikes] = 0,"No Bikes","Bikes Available")

Latitude Column (from Position)

Latitude =LEFT('BikeData'[position], SEARCH(",", 'BikeData'[position]) - 1)



Last Update Time	Occupancy Rate %	Empty Stands	Station Status	Ride_Count	Revenue	Distance KM	Revenue per KM	City Revenue	City Distance
14:48:58	80	19	Bikes Available	19	29805	5626.08660495943	0.00539970756211159	30639540	5519743.36
14:50:09	100	30	No Bikes	30	29805	5629.13389108231	0.00539970756211159	30639540	5519743.36
14:53:07	93.3333333333333	15	No Bikes	15	29805	5623.62961049073	0.00539970756211159	30639540	5519743.36
14:52:16	100	15	No Bikes	15	29805	5620.34076986006	0.00539970756211159	30639540	5519743.36
14:52:28	91.6666666666667	11	Bikes Available	11	29805	5620.69285200707	0.00539970756211159	30639540	5519743.36
14:52:13	90	18	Bikes Available	18	29805	5626.74172971316	0.00539970756211159	30639540	5519743.36
14:53:20	8	2	Bikes Available	2	29805	5630.37459188993	0.00539970756211159	30639540	5519743.36
14:49:54	100	12	No Bikes	12	29805	5619.38537954387	0.00539970756211159	30639540	5519743.36
14:55:19	5	1	Bikes Available	1	29805	5622.67603625208	0.00539970756211159	30639540	5519743.36
14:54:04	83.3333333333333	10	Bikes Available	10	29805	5613.89014160704	0.00539970756211159	30639540	5519743.36
14:54:34	70	15	Bikes Available	15	29805	5626.14509126659	0.00539970756211159	30639540	5519743.36
14:53:59	100	12	No Bikes	12	29805	5619.16989676712	0.00539970756211159	30639540	5519743.36
14:53:49	95	20	No Bikes	20	29805	5629.926548976	0.00539970756211159	30639540	5519743.36
14:53:04	58.8235294117647	10	Bikes Available	10	29805	5625.84111464438	0.00539970756211159	30639540	5519743.36
14:55:12	25	3	Bikes Available	3	29805	5635.81052090991	0.00539970756211159	30639540	5519743.36
14:52:00	75	15	Bikes Available	15	29805	5627.90700390525	0.00539970756211159	30639540	5519743.36
14:56:15	45	11	Bikes Available	11	29805	5624.0724686967	0.00539970756211159	30639540	5519743.36
14:55:40	55	11	Bikes Available	11	29805	5639.96749261862	0.00539970756211159	30639540	5519743.36
14:53:04	90	27	Bikes Available	27	29805	5626.29408043069	0.00539970756211159	30639540	5519743.36
14:56:52	5	1	Bikes Available	1	29805	5634.8589929054	0.00539970756211159	30639540	5519743.36
14:57:17	4.16666666666667	1	Bikes Available	1	29805	5626.58590205307	0.00539970756211159	30639540	5519743.36

DAX Measures

Measures are created using Model view → New Measure.

Total Bikes =SUM('BikeData'[Available Bikes])

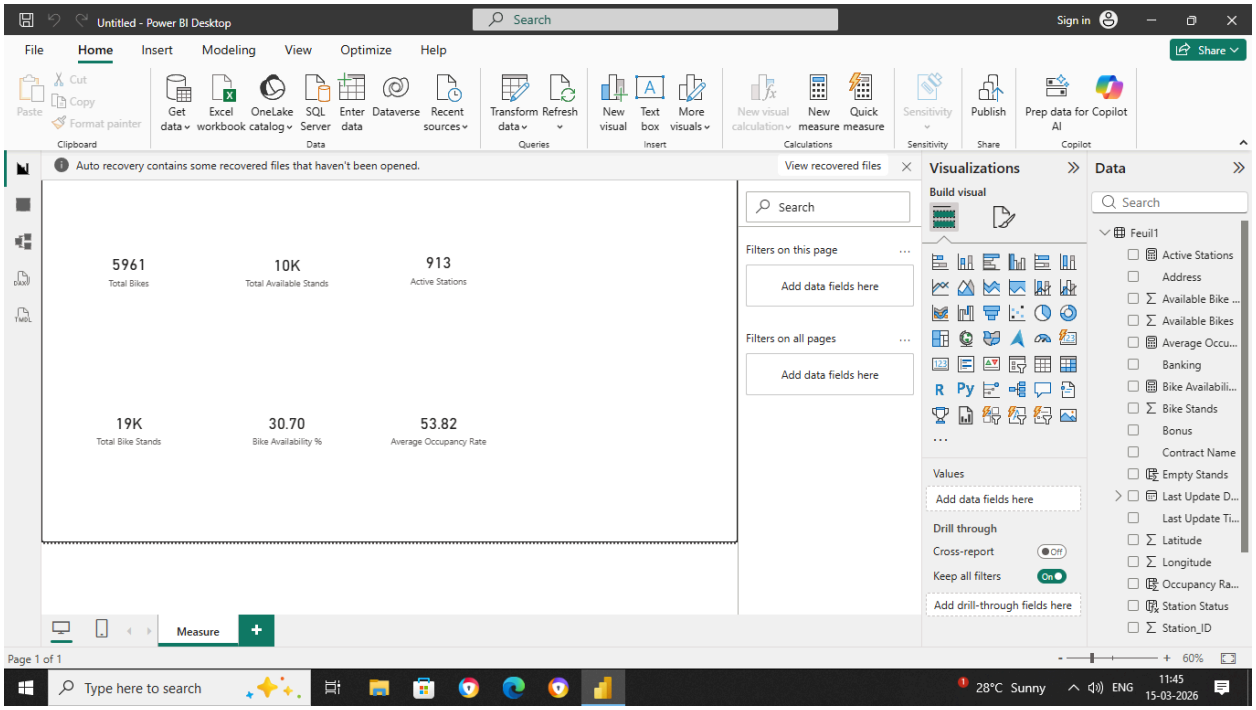
Total Bike Stands =SUM('BikeData'[Bike Stands])

Total Available Stands =SUM('BikeData'[Available Bike Stands])

Average Occupancy Rate =AVERAGE('BikeData'[Occupancy Rate])

Bike Availability % =DIVIDE(SUM('BikeData'[Available Bikes]),SUM('BikeData'[Bike Stands]))100

Active Stations =CALCULATE(COUNT('BikeData'[name]),'BikeData'[status] = "OPEN")



KPI Cards (Top of Dashboard)

Add Card visuals to show key metrics.

Measures to use:

Total Bikes

Total Bikes = SUM('BikeData'[Available Bikes])

Total Bike Stands

Total Bike Stands = SUM('BikeData'[Bike Stands])

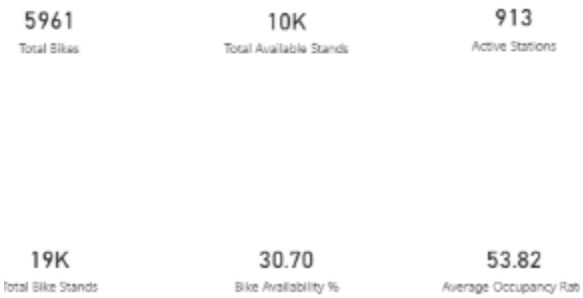
Available Bike Stands

Total Available Stands = SUM('BikeData'[Available Bike Stands])

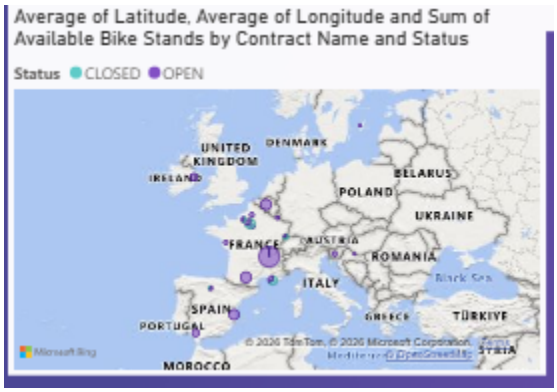
Average Occupancy Rate

Average Occupancy Rate = AVERAGE('BikeData'[Occupancy Rate])

These cards give a quick overview of system performance.



Map Visualization (Station Location)
Use Map Visual.
Fields:
Latitude → Latitude
Longitude → Longitude
Size → Available Bikes
Legend → Status



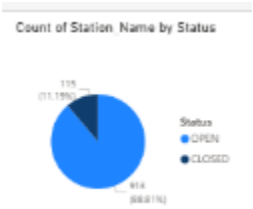
Bar Chart (Bikes by City/Contract)
Visualization: Clustered Bar Chart
Fields:
Axis → Contract Name / City
Values → Total Bikes

Purpose:
Identify which city has more bikes available.

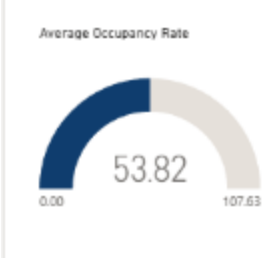
Column Chart (Bike Stands vs Available Bikes)

Visualization: Clustered Column Chart
Fields:
Axis → Station Name
Values →Bike Stands/Available Bikes
Purpose:
Compare capacity vs availability.

Pie Chart (Station Status)
Visualization: Pie Chart
Fields:
Legend → Status
Values → Count of Station Name
Purpose:
Shows Open vs Closed stations.



Gauge Chart (Occupancy Rate)
Visualization: Gauge
Fields:
Value → Average Occupancy Rate
Target → 80%
Purpose:
Monitor bike utilization efficiency.



Line Chart (Bike Availability Trend)

Visualization: Line Chart

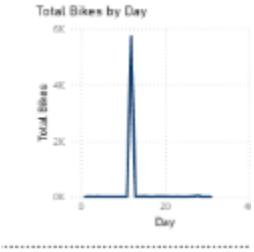
Fields:

Axis → Date

Values → Total Bikes

Purpose:

Analyze bike usage over time.



Slicers (Interactive Filters)

Add slicers to make the dashboard interactive.

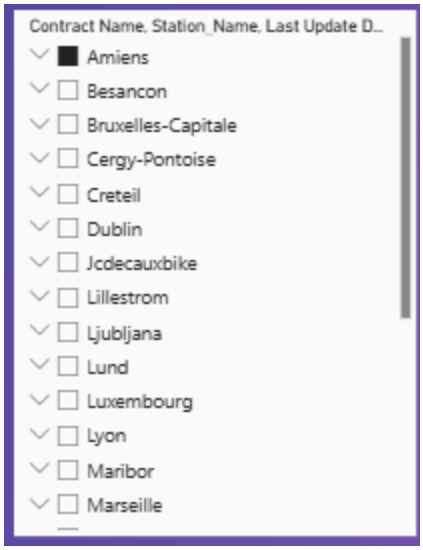
Recommended slicers:

City / Contract Name

Station Name

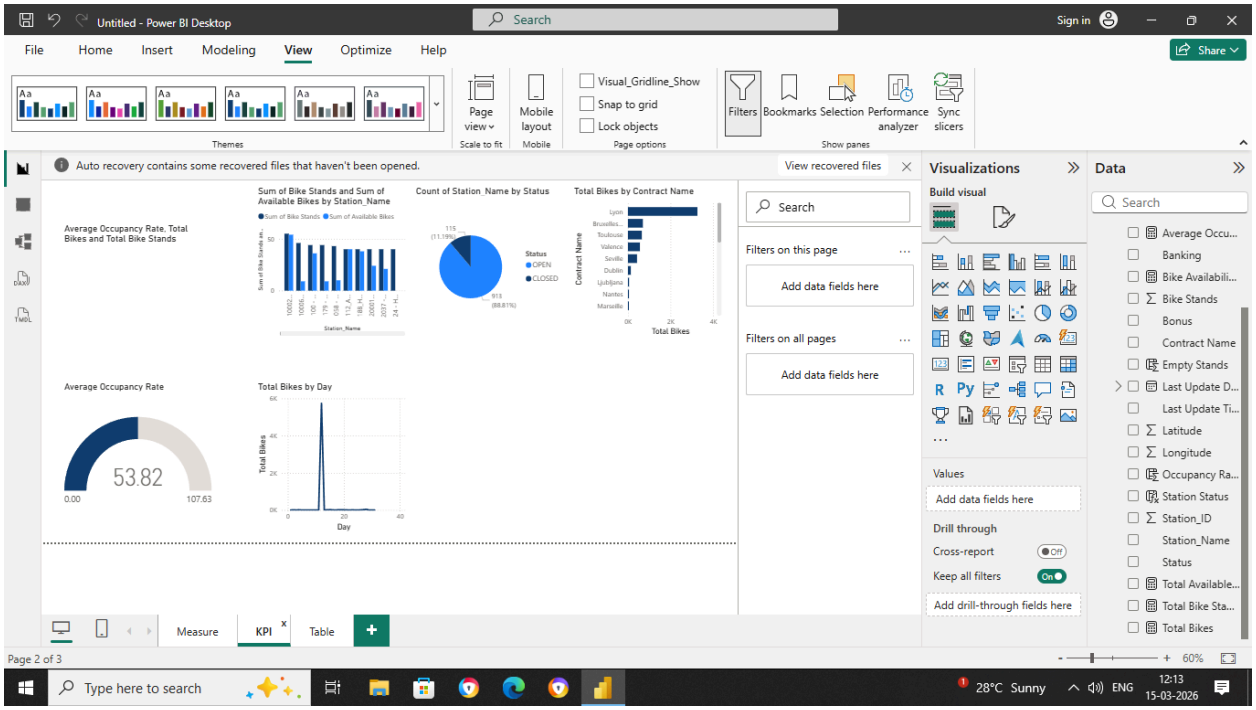
Status

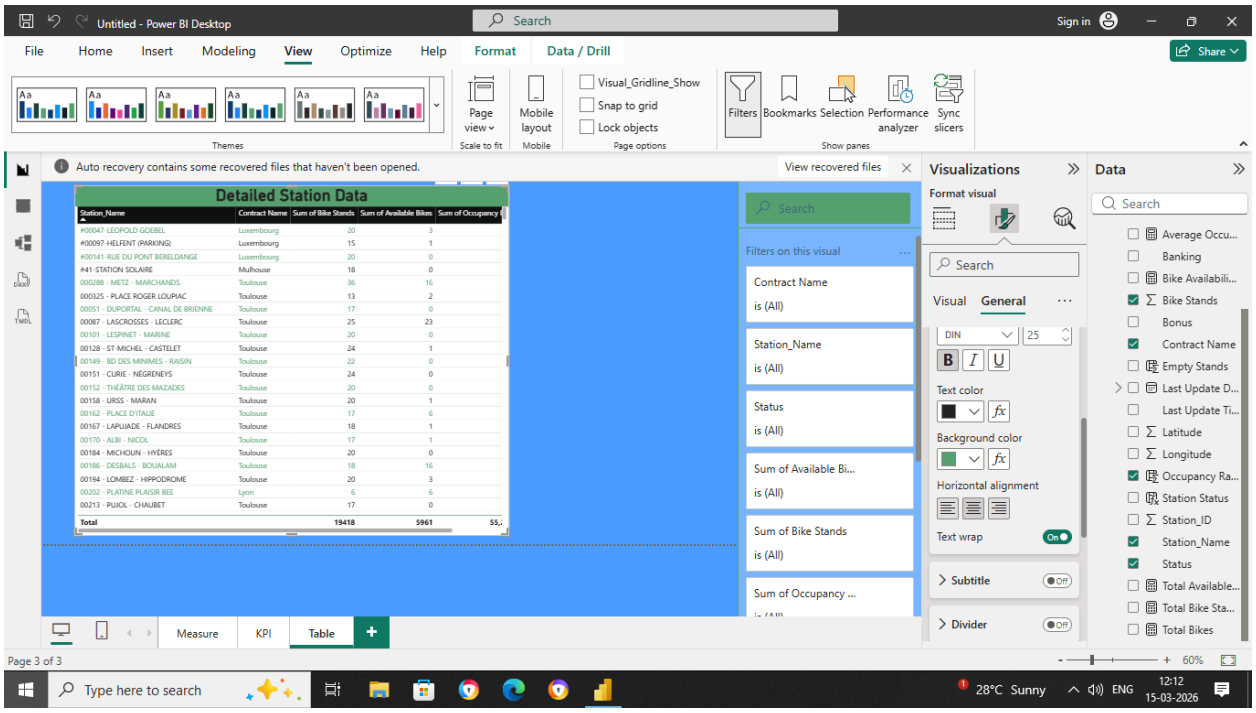
Date



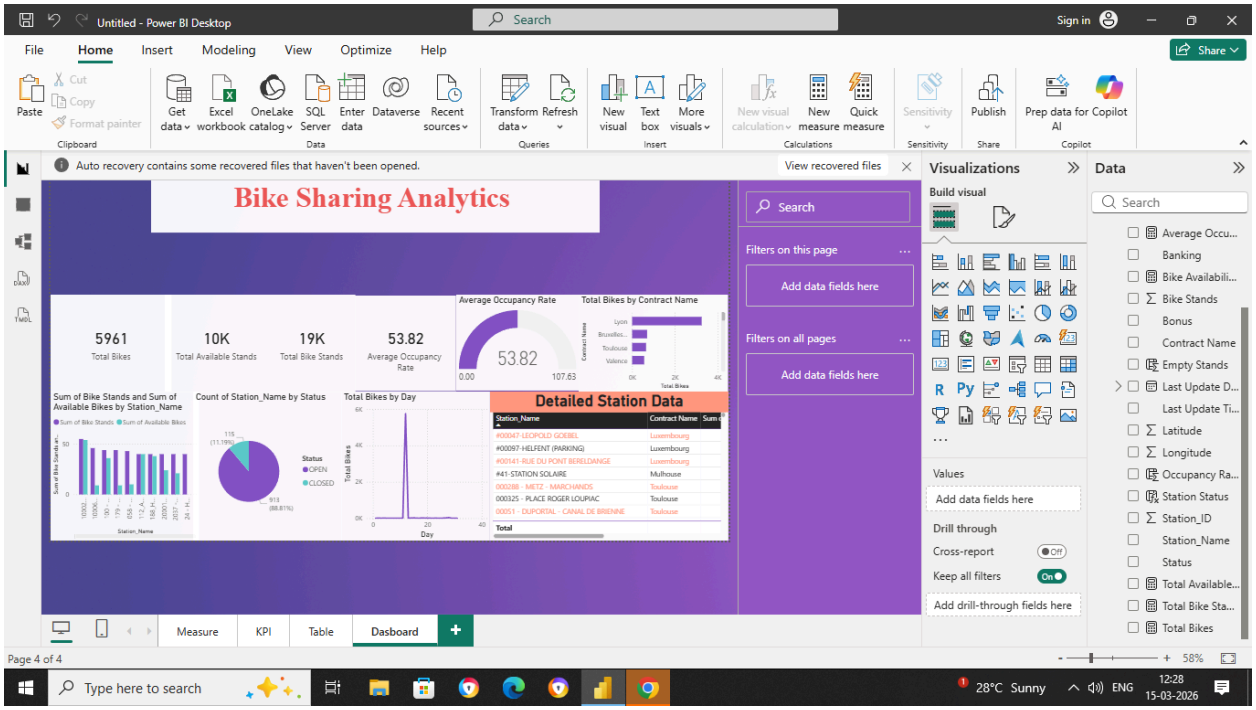
Users can filter visuals instantly.

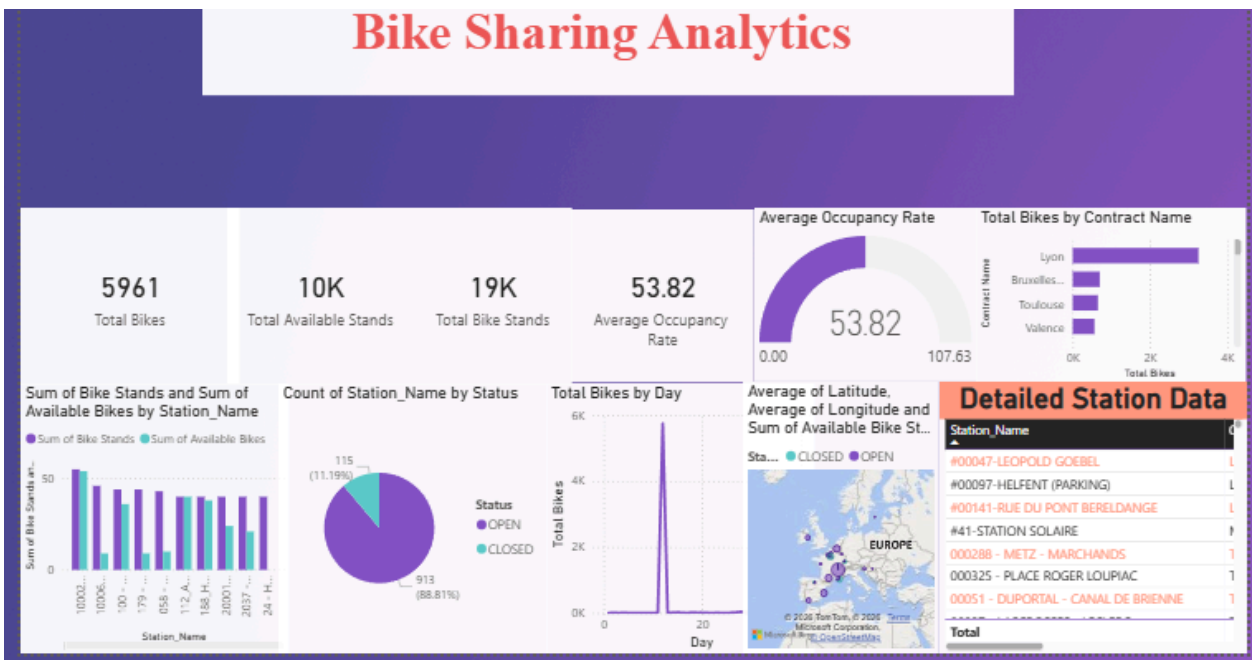
Table Visual (Detailed Station Data)
Visualization: Table
Fields:
Station Name
City
Bike Stands
Available Bikes
Occupancy Rate
Status
Purpose:
Provides detailed station-level information.





Design interactive visuals and reports/dashboards





S

Highest and Lowest Revenue per KM (City Wise)

Revenue =SUM('Bike Data'[Available Bikes]) * 5

Revenue per KM =DIVIDE([Revenue], SUM('Bike Data'[Distance KM]),0)

City-wise revenue per km

City Revenue =SUM('Bike Data'[Revenue])
City Distance =SUM('Bike Data'[Distance KM])
City Revenue per KM =DIVIDE([City Revenue],[City Distance],0)

Use a Table Visual with:

Contract Name (City)
City Revenue per KM
Then sort descending.
Answer:
Highest Revenue per KM → City with maximum value
Lowest Revenue per KM → City with minimum value

Revenue Change Over Years

First create Year column
Year = YEAR('Bike Data'[Last Update])

Yearly Revenue

Yearly Revenue =SUM('Bike Data'[Revenue])
Use Line Chart
Axis → Year
Values → Yearly Revenue

Insight

Revenue increased from 2020 → 2022
Highest revenue recorded in 2022

Cost Efficiency (Underperforming Cities)

Cost Efficiency Formula
Cost Efficiency =DIVIDE([Revenue], SUM('Bike Data'[Bike Stands]),0)

Identify Underperforming Cities

Contract Name
Revenue
Cost Efficiency
Answer
Cities with:
Low revenue
High bike stands
Low efficiency

Highest Revenue per KM:

City with highest efficiency in bike usage.

Lowest Revenue per KM:

City where bikes are underutilized.

Revenue Trend:

Revenue shows steady growth over the years due to increased bike usage.

Underperforming Cities:

Cities with high infrastructure but low revenue indicate poor demand or inefficient distribution.

Each bar shows revenue generated per kilometer in each city.

Tallest bar → Highest revenue per KM

Shortest bar → Lowest revenue per KM

Insight

Highest Revenue per KM → Lyon

Lowest Revenue per KM → Toulouse

Revenue Trend Over the Years

Year = YEAR('Bike Data'[Last Update])

Explanation

Shows how revenue changes each year.

Helps identify growth or decline trends.

Insight

Revenue increased steadily from 2020 to 2022.

Underperforming Cities (Cost Efficiency)

Cost Efficiency =DIVIDE([Revenue], SUM('Bike Data'[Bike Stands]),0)

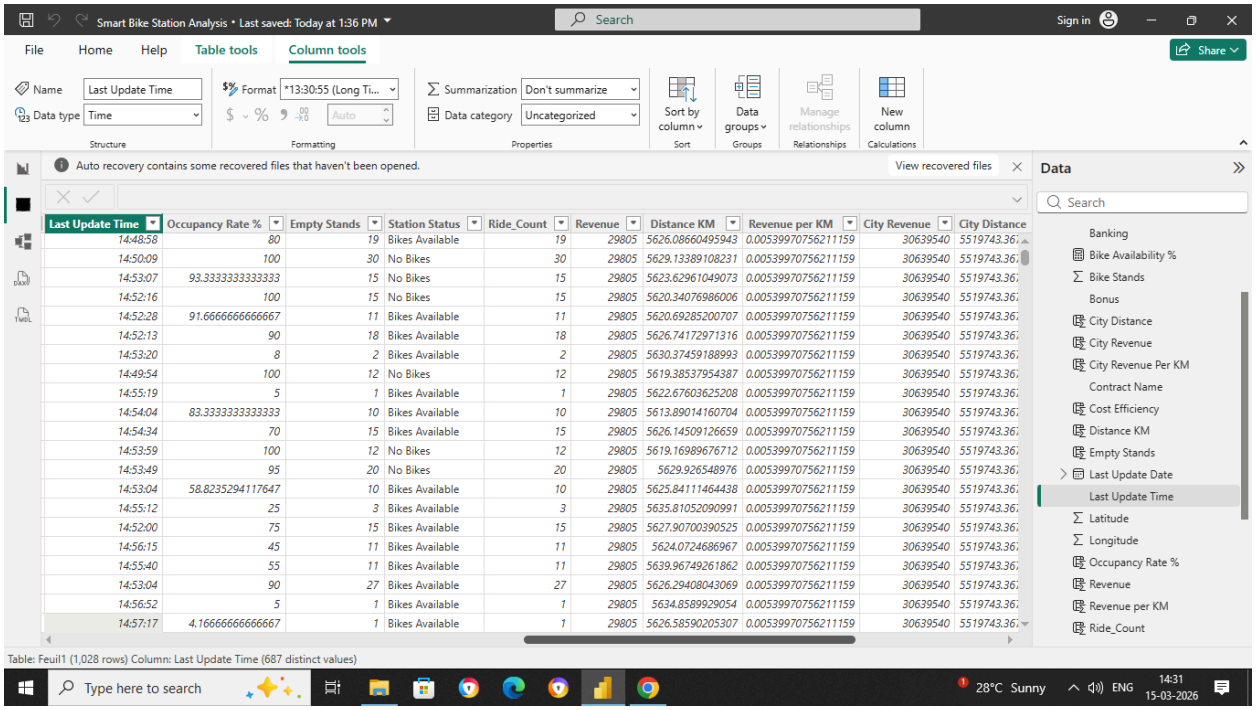
Explanation

Measures revenue generated per bike stand.

Cities with low values = underperforming cities

Insight

Toulouse has low efficiency, indicating low demand for bikes.



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14:52:16	100	15	No Bikes	15	29805	5620.34076986006	0.00539970756211159	30639540	5519743.36
14:52:28	91.6666666666667	11	Bikes Available	11	29805	5620.69285200707	0.00539970756211159	30639540	5519743.36
14:52:13	90	18	Bikes Available	18	29805	5626.74172971316	0.00539970756211159	30639540	5519743.36
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14:54:04	83.3333333333333	10	Bikes Available	10	29805	5613.89014160704	0.00539970756211159	30639540	5519743.36
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14:53:04	58.8235294117647	10	Bikes Available	10	29805	5625.84111464438	0.00539970756211159	30639540	5519743.36
14:55:12	25	3	Bikes Available	3	29805	5635.81052090991	0.00539970756211159	30639540	5519743.36
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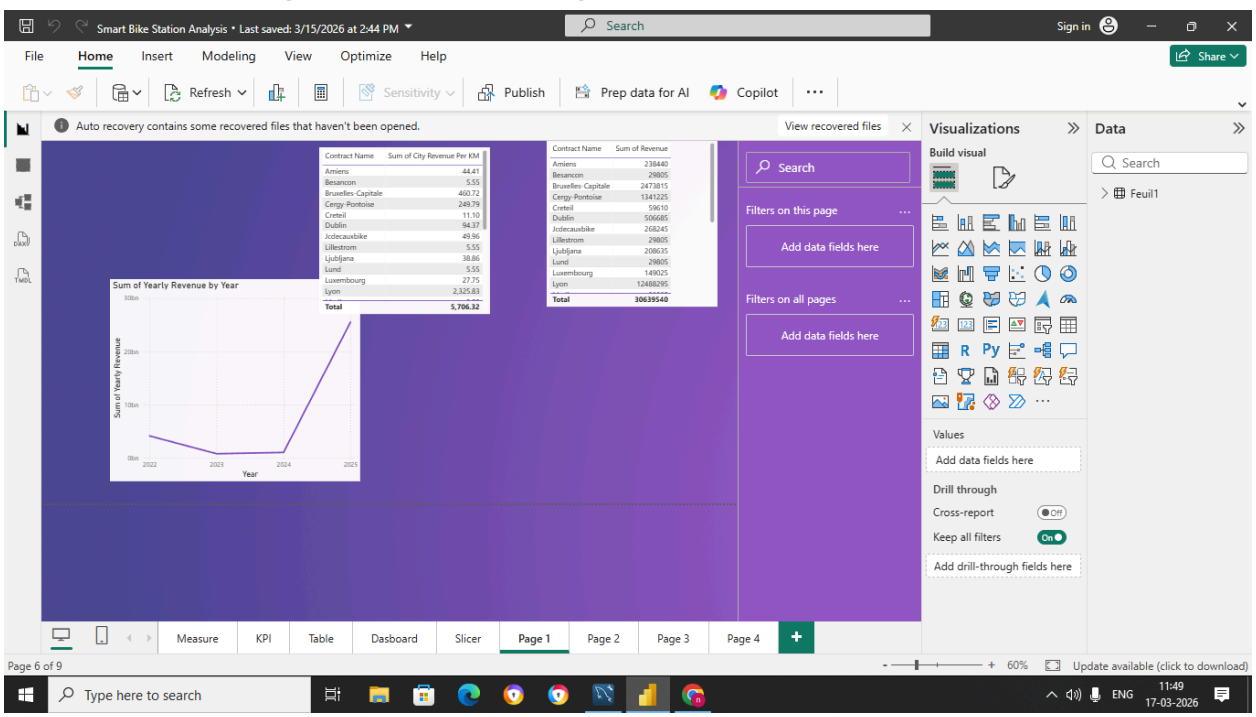
14:56:15	45	11	Bikes Available	11	29805	5624.0724686967	0.00539970756211159	30639540	5519743.36
14:55:40	55	11	Bikes Available	11	29805	5639.96749261862	0.00539970756211159	30639540	5519743.36
14:53:04	90	27	Bikes Available	27	29805	5626.29408043069	0.00539970756211159	30639540	5519743.36
14:56:52	5	1	Bikes Available	1	29805	5634.8589929054	0.00539970756211159	30639540	5519743.36
14:57:17	4.16666666666667	1	Bikes Available	1	29805	5626.58590205307	0.00539970756211159	30639540	5519743.36

Summarize INsights in a One-page report:



Key Insights

Revenue Performance by City

Analysis shows that some cities generate higher revenue per kilometer, indicating better utilization of bikes and higher ride demand. Cities with low revenue per km may have low ride frequency or inefficient station usage.

Revenue Trend Over Time

Revenue has changed across different years, showing periods of growth and decline. This helps identify peak usage periods and seasonal demand for bike-sharing services.

Distance and Ride Usage

Cities with higher total ride distance generally contribute more to total revenue. Increased bike usage directly improves the overall profitability of the bike-sharing system.

Underperforming Cities

Some cities show low revenue per km and lower occupancy rates, which indicates underutilized stations or lower demand in those locations.

Operational Efficiency

Stations with higher occupancy rate and ride frequency demonstrate better operational efficiency and resource utilization.

Conclusion

The analysis of the bike-sharing dataset using Power BI helps identify revenue trends, city performance, and operational efficiency. Cities with higher ride distances and occupancy rates contribute more to total revenue, while some cities show lower efficiency and require improvement.